



Therapeutic Education Among Cancer Patients

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Abstract

Therapeutic education is one of the most important educational methods that play a positive role among the patient, especially the cancer patient. It is the means that helps the patient overcome his illness and reduce its severity, and its availability in the various centers that receive patients helps reduce the costs of treatment and coexistence with the disease.

This article aimed to introduce this method (therapeutic education), by identifying its concept, importance, how to conduct it and its recommendations, in general and with cancer patients in particular.

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1. Introduction

The life changes acceleration of the individual, in various aspects, whether social or economic, clearly affected his lifestyle. Which led him to follow and adopt behaviors that most of them pose a real and clear danger to his health. Behavioral risk factors such as lack of physical activity, frequent intake of unhealthy foods, continuous exposure to radioactive materials, indiscriminate drug use and other risky behaviors, in addition to persistent or severe exposure to psychological problems such as chronic stress, depression and anxiety. It has a major role in changing the balanced chemical system of the organism, and weakening the body's immunity. This is what makes the individual vulnerable to various chronic and terminal diseases, for example cancer, which is increasing in incidence.

Due to this noticeable and frightening increase, Algeria tried to provide various material and human capabilities. As well as to address this terminal disease by providing preventive and therapeutic strategies appropriate with the disease, through the establishment of cancer control centers at the level of various health institutions across the country. In addition to the formation of medical and paramedical staff based on patient treatment, developing a program of training in psycho-oncology for the psychologists working in these centers in order to help the patient fight cancer. Dealing with the various problems and effects of the disease, whether physical, psychological, social or spiritual from the moment the diagnosis is announced.

Thus, all support is directed towards creating programs that help improve patient care, such as therapeutic education. Which is concerned with teaching patients the physiological foundations of their disease, as well as preventive and curative skills, to reach self-management. Rather, it goes beyond it to interfere in the formation of the personality, which is affected by the self-image of the individual, his field of interests and his wide knowledge of the chronic disease itself and its physical and psychological side effects.

A number of researchers have pointed out that individuals who suffer from organic diseases have the ability to organize cognitive about health and disease through which they explain and understand their pathological symptoms and through which their reactions to the disease also appear (Mewefek, 2017). Which makes the design and implementation of therapeutic education programs for cancer a matter that contributes a great role in helping them overcome the various difficulties and problems that they face.

1.0. Problem statement :

All diseases kinds, whether they are chronic, terminal or endemic, are among the most important problems facing the individual in various societies. So that it is currently attracting the attention of various studies and researches, by occupying priority among the main causes of disability and death. Among the most dangerous and life-threatening diseases is cancer. This is due to the effects it leaves on the various aspects of the patient's life (psychological, social, physical, family...

and others), which in turn affects his well-being.

Based on the statistics conducted by the World Health Organization (2019), it was found that cancer is the second cause of death globally, after heart disease. It affects more than 14 million people annually. This number is expected to reach more than 21 million by 2030. It also causes about 8.8 million deaths annually, and the International Agency for Research on Cancer estimated in 2018 a million deaths worldwide. An estimated 44 million people are alive within 5 years of being diagnosed with the disease. This is called the five-year survival rate (Who, 2019).

Algeria records huge numbers of cancer. According to the numbers presented by the fourth annual meeting of the Algerian Network for the Cancer Registrars in 2018, the increase rate has reached 50% in the last ten years, ranking second with a rate of up to 21% in the death causes of Algerians after heart and arteries diseases. And the number of people infected with this disease reached 42,700 patients at the end of 2018, with an annual increase of 7%, at a rate of 103 cases per 100,000 people (Bourenan, 2018).

This, cancer can be considered one of the diseases that most involve disability, which occurs as a result of the disease complications or associated treatments, which requires motivation and training for the patient in order to take care of himself. Therefore, it was necessary to search for solutions that would reduce the severity of the disease and its consequences and help the patient to improve, and

therapeutic education is considered one of them (solutions).

Therapeutic education aims to reach balance and learn the necessary skills and abilities to coexist with the disease and improve the patient's lifestyle, regardless of his age. This is in the form of private individual, and group models (Who, 1998). As a means that allows collecting various information related to the disease, how it develops, and its treatment program. As well as how to prevent and limit the development of the disease, within the knowledge and education, knowledge relates to an individual's regulation of time, energy, memory, identity, and development (Calops et al, 2008). Acquiring patient a set of competencies related to dealing with his disease and others related to himself and increasing his value, which determine his attitudes towards a better self esteem, his disease and his surroundings. Degennaro Louis explained this in a letter on how to live with cancer in children. Where he explained a set of steps linking between risk factors and the disease prognosis that establishes the steps for designing a therapeutic education program (Degennaro, 2018).

In this regard, Algeria tried to adopt a new strategy that focuses mainly on the patient through the National Plan for Cancer Control (2015-2019). As one of its goals was to prioritize therapeutic education in order to improve the quality of life of patients undergoing treatment. As one of its goals was to prioritize therapeutic education in order to improve the quality of life among patients undergoing treatment (Plan National For Cancer). Despite the efforts made in this field, confusion still

exists between the concepts of therapeutic education and health education. Most individuals and specialists do not differentiate between the two concepts and consider them to be one concept, but this is a common mistake. Health education is a comprehensive concept concerned with prevention and the improvement of the general health of societies. In contrast to therapeutic education, whose concept is limited to the presence of disease, regardless of its type, chronic or fatal. Hence, this study aimed to clarify the nature of therapeutic education, as well as its objectives, benefits, and the foundations on which it is based, the profile of the person with cancer, and finally, what is the therapeutic education for cancer patients ?

1.1. *The study importance :*

Undoubtedly, everyone knows the importance attached by most of the official and unofficial authorities and institutions in our day to all kinds of diseases spread in our society, this is due to the unreasonable number of diseases. The importance of our study are :

_ Increasing the attention of those in charge of the official institutions in the sector to the necessity of adopting various programs that help patients to resist their disease and deal with it effectively by adopting therapeutic education programmes.

_ Addressing the concept of therapeutic education, which is one of the modern concepts in our current era.

_ To illustrate the importance of applying therapeutic education programs by reducing the exorbitant costs of health care.

1.2. *The study objective :*

This study aimed

-To identify the concept of each of the therapeutic education, its importance, the necessary steps to apply it, and its recommendations.

-To determine the profile of the target personality for cancer, as well as identifying the personality type "C" and the incidence of cancer.

-To the nature of therapeutic education for cancer patients.

2. *Litrature review :*

2.1. *Cancer :*

Cancer is one of the most prevalent chronic diseases today. It is one of the biggest challenges facing the health systems in the world, as it is the second leading cause of death after heart disease. It is a serious, life-threatening disease, which often appears in different forms according to injury, and usually leaves serious physical and psychological effects and consequences.

Cancer is defined as the continuous, unorganized or untamed division of cells, and this division leads to the formation of a huge number of cells. It depends on the genetic side of the individual, or depends on his environment and lifestyle in general (Mohamed Al djisi, 2020).

2.2. *Kinds of cancer and its causes :*

Cancer has many kinds, which vary relatively to gender, as there are certain kinds of this disease that affect women because the structure of the female body differs from that of the male, for example: uterine, breast, or ovarian cancer. There are

kinds of this disease that affect males more than women, such as lung cancer, because the majority of smokers are men if we link the disease to the cause of smoking. There are kinds of disease that affect both sexes in a relatively close.

On the other hand, there are factors that cause the emergence of cancerous diseases, including environmental factors such as chemicals used in industries and technology. Their increased use is a major cause of cancer. This has been proven in scientific research, and genetic factors related to the family genetic structure or the group that carries inherited syndromes that are associated with an increased incidence of cancerous tumors, or psychological factors such as psychological stress, anxiety, etc. In addition to the presence of occupational factors effect, as there are professions that help in a certain kinds of cancer incidence, and the habits of dietary factor that differ from one individual to another (Abed Alrazek, 2013).

2.3. The profile of targeted personal for cancer :

2.3.1 Definition of personality type (c)

This type appeared by Grèer and Morris (1980), where they defined it as the character that appears with a loving and understanding appearance. However, It achieves this appearance by using a very large defensive effort, and therefore it is not as adaptive as it appears. In the face of stressor factor, these individuals show greater autonomy stimulate than others. Such individuals get low scores when we measure their negative emotions using auto-evaluation method and grades

are raised when distress is measured by observation and conversation.

Temoshok defined it in 1987 as a cognitive-behavioral pattern of adjustment with stressful social situations full of conflict and threatened with loss, as adjustment is characterized by (the emotion cancel and a sense of helplessness and despair) and fluctuate between exaggerated high adaptation and a sense of helplessness associated with despair and depression.

2.3.2 Characteristics of personality type "C":

Type C personality, according to Temoshok and colleagues, is characterized by the following set of traits

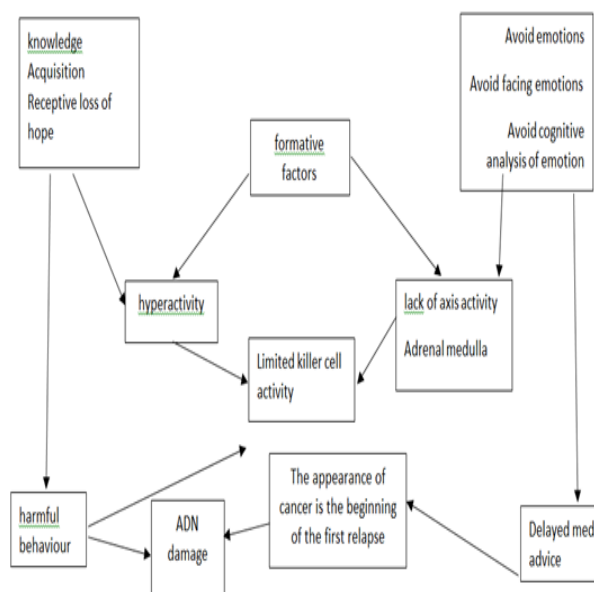
- 1- Inability to express anger and release tension.
- 2- The tendency to agree and obey.
- 3- Uncertain self-sacrifice, inclined to indulgence and avoid conflict.
- 4- Patient and calm, able to wait and accept time pressure.
- 5- He does not disclose his emotions.
- 6- Low self-esteem (sees himself as insignificant, inept, incompetent, and chaotic).
- 7- Adhering to routine, living in psychological loneliness.
- 8- He suffers from depression and pessimism and feels hopeless.

2.3.3 Personality type C and cancer :

In the study of Grèer and Morris (1979) it was mentioned that cancer patients are characterized by some common traits (depressive knowledge,

hopelessness, surrender, and emotional inhibition, and the study of Uirshing (1982) concluded that there is a relationship between personality type "C" and submission, surrender and the incidence of breast cancer. Contrada (1990) concluded that personality type "C" is associated with cancer. Hence inhibition and suppression of emotions are significantly present among people with cancer. The following a chart showing the mechanism of cancer occurrence.

Figure N° 1: The Centrada model explains the incidence of cancer (1990).



Source: Belkhir, (2020)

2.4. Therapeutic Education :

Undoubtedly, each topic has a theoretical literature that talks about and clarifies it. In the beginning, before defining therapeutic education, In the beginning, before defining therapeutic education, It is necessary to refer to a concept that is close in use to the concept of therapeutic education. This is to clarify the difference

and confusion. This concept is health education, which is a comprehensive concept.

2.4.1. Definition of health education :

Health education facilitates the process of modifying healthy behaviors, and it can be defined as follows :

The World Health Organization (1998) : defined it as opportunities to build awareness of learning involving some forms of communication aimed at improving health knowledge, including improving knowledge, developing life skills that help maintain individual and community health.

In another definition of the World Health Organization, : Health education is considered the most powerful weapon for public health, it represents an important health field that works to achieve the concept of public health by spreading health awareness and health culture, and in fact it seeks to provide individuals with correct health information it urges them to adopt positive healthy behaviors, and it also provides children with decision-making skills in different life and health situations (Yakhlef, 2001)

2.4.2. Defination of therapeutic education :

The therapeutic education of a patient with a chronic disease can be considered a continuous process, and integrated care that focuses on and cares for the patient, and also includes organized activities aimed to spreading awareness and information to reach learning and psychological and social support related to the disease, as well as prescribed treatment, care, hospitalization or other health care

institutions related to health behaviors (Vervloet, 2002).

A process that works to help the patient and his family to understand the disease and treatments, participate in the treatment, take care of the patient's health condition, and encourage him to return to daily activities and life projects. This process implements by (associations, therapists, patients and their families). Its goals related to health defined through objective and subjective needs, implemented within a formal and systemic framework through the integration of resources, and uses pedagogical methods and means, and requires competencies and coordinating structures (Ficher, 2006).

It was also defined by the European World Health Organization during the year 1996 as "working to help the patient maintain and retention the skills he needs, in order to better manage his life associated with a chronic disease." It basically emerges as a process for taking care of the patient, and it contains organized activities related to psychological and social support in order to make the patient aware of his illness and treatment, the process of hospitalization operations, behaviors related to health and disease, and the nature of his treatment through coordination so that each one bears its responsibility to reach the goal of helping him to improve his quality of life (WHO, 1996).

The therapeutic education for cancer patients can be considered educational and special, it affects both the family, the patient, also the role of the treatment team. This tripartite relationship makes us reach results that make the patient improve and

overcome his pain, according to defining a healthy and preventive lifestyle.

2.4.3. The significance of therapeutic education :

Therapeutic education has a great and clear importance that is used to the patient health, emotion, people around him and care givers. As it increases and affects the individual, for example, the patient's confidence increases, he becomes more accepting of himself and others and the patient's treatment adherence (De Broca, 2006).

Mastering and learning how to cope with the disease and manage it through the continuity of self-care and medical follow-up, reducing the severity of symptoms, acquiring and learning knowledge management and how to solve problems by controlling the progression of the disease, increase the patient's independence in carrying out daily activities by facilitating adherence to prescribed treatments and improving his quality of life. Learn to make plans based on the patient's participation in putting a plan for the therapeutic education program (Bacoran, 2006), Improving and developing the relationship between the doctor, the patient, his family, and the health care provider. Healthy behaviours education (HAS, 2007), Taking responsibility, whatever it may be (his responsibilities in health care), The failure of respect the treatment in its various periods, reducing it, leaving it, and not regularizing it may result in additional direct or indirect economic costs (Benrebrit & Rahmoun, 2016).

To sum, the therapeutic education plays an important role in patient life, regardless the type of his illness and whatever his age.

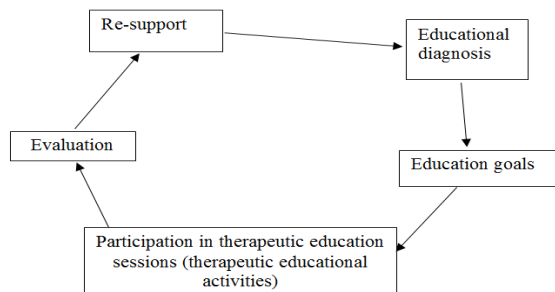
They are used in different and uneven ways, and if patient follows its steps, he will inevitably reach a positive result, and vice versa. In the following the explaining of Steps to apply therapeutic education.

2.4.4. Steps to apply therapeutic education :

Undoubtly, the therapeutic education is a necessary means in patients life, and it is like any other means requires steps to be applied. The steps for applying therapeutic education can be summarized as follows :

Educational diagnosis, negotiated contract, educational sessions, and evaluation, and this relates to a model developed by J-Fd'Ivernois and R Garmayre (Kremo & Roussey, 2010)

Figure N° 2: Shows the therapeutic education cycle



Source: Baud , (2009)

2.4.4.1. Educational diagnosis in patient :

The first step in the educational process is the educational diagnosis, and it is considered one of the steps that gradually develop, and we can do the educational diagnosis during the first consultation and finish it during education, taking into account its re-evaluation in each meeting with patient.

Through an individual and structured interview is collected information related to patient's personality, his knowledge of the disease, his requests, his capabilities, his motives in knowing and managing his disease and conditions, even with regard to his life, his personal and professional project. Where in this way it is possible to identify the skills that patient is intended to acquire them through therapeutic education, and priority must be given to these skills according to their importance among patient and the extent of his acceptance of them (SRRP , 2001).

Practically, the educational diagnosis is represented in understanding what the patient understands and how to live with his disease and treatment. It is also represented in giving a score of his mastery of observation, treatment techniques measures and evaluating his gesture capabilities. It is also being estimated and evaluated his abilities in decision-making in adapting his treatment and facing various accidents (Simon et al, 2013). The educational diagnosis dimensions are:

2.4.4.1.1. The biomedical dimension of disease :

It means what patient has, and its aim is try to know patient's medical history, including the duration, development, and severity of disease, as well as other health problems that are important to patient, the reasons for hospitalization, and others.

2.4.4.1.2. The social and professional dimension :

It means what patient does and its goal related to the lifestyle : daily life, leisure time, the social, family and professional environment, and so on.

2.4.4.1.3. Cognitive dimension :

It means by it what patient knows about his disease and it concludes patient's beliefs about the disease, which is represented in the patient's knowledge of his disease previously, as well as his beliefs, concepts about the mechanisms of the disease, and the factors causing it, the effectiveness of treatment, and so on.

2.4.4.1.4. The emotional-psychological dimension :

The emotional-psychological dimension : It means who the sick person is and it includes several followed stages of the accepting disease process, they are as follow, the initial shock, followed by denial, rebellion, compromise, depression, and finally acceptance of disease.

2.4.4.1.5. The patient's project:

means what their project is and its content represented the definition of patient to his initial project, followed by presenting the project as achievable through education.

2.4.4.2. Agreed contract:

It is a process of negotiating goals with the aim of achieving them between the examiner and the subject, and it is the second step in therapeutic education (Kremp & Roussey, 2010). It represented in signing a partnership contract that obliges all the participating sides to abide by everything related to this agreement,

whether it is related to the objectives of the educational diagnostic sessions that are related to the skills that the patient must acquire (SRRP, 2001). With a joint planning process in which the various educational, health and therapeutic goals are outlined, this step is primarily a management process.

In sum, while developing a partnership between the health care provider and the cancer patient, all points related to the conduct of the various sessions must be clarified in order for the program to be successful and beneficial to the patient and to have a positive effect on the patient.

2.4.4.3. Participation in therapeutic education sessions :

It includes a set of individual sessions or in general group examinations of about 45 minutes, through which the patient becomes gradually aware of the skills he acquires. Technology is used in a different ways that and adapted to diseases concerned with the activation of learning techniques and methods, for example (scenarios, pictures, videos, competition, observation...) (Piperini , 2016). Thus, it can be integrated into learning this is during educational courses that allow the patient to realize his successes, difficulties and mistakes, and what develops his self-evaluation. So, he can carry out automatic learning activities and check whether all the skills described in the contract have been acquired on them or not through the activities of the planned learning sessions (SRRP, 2001).

2.4.4.4. Evaluation :

At this stage, the progress of patient knowledge is checked and its development, improve care behavior, with the positive development of medical data (Piperini, 2016). It is also an opportunity to adapt educational therapy in the background and treatment plan in case of any exacerbation (SRRP , 2001) This stage depends on 03 evaluative dimensions, which are indicators of learning, and they are: bringing about a change in behaviors in a healthy and proper manner, and continuity over time (commitment to these new behaviors to become a habit within the lifestyle) and experience (Schunk , 2012).

Therefore, the evaluation focuses on the skills acquired by the patient from curricula, methods, and educational skills for caregivers. For example : it can evaluate skills acquired by cancer patients gradually and objectively, according to the goal. It can also evaluate them individual or group at the end of the program (HAS, 2007).

2.4.5. Recommendations for the application of therapeutic education :

Therapeutic education is an educational process that focuses specifically on the patient, as follows :

Coping strategies, health locus of control, beliefs and representations about health, subjective and objective needs related to the patient, declared or hidden, must be related to the therapeutic process. It is directly interested in the daily life of the patient and his surroundings, whether on the psychological or social level, and works to involve families as well, close relatives

and friends of the patient. It evaluates the patient's health condition and adapts it according to disease prognosis and his lifestyle on lifespan. It should be sequential, organized, and implemented through a variety of educational capabilities, It considered multidisciplinary, It applied within a health network, and It applied by health professionals who are trained in the field of therapeutic education (WHO.1998)

2.4.6. Therapeutic education among cancer patients :

Therapeutic education for cancer patients or in oncology is a term that has recently emerged, unlike other chronic diseases such as diabetes and asthma. This was confirmed by most studies, as the study of Gallefoss.F, Bakker.P.S, Kjoersgaard.P in (1999) entitled by the Quality of Life Evaluation after Educating the Patient to Control Asthma and Chronic Obstructive Pulmonary Disease, a randomized study applied a therapeutic education program distributed over group sessions and an individual session for each nurse and physiotherapist and self-management, the result was that asthma patients gained self-management and quality of life (Chegrouna, 2021). As for diabetes, several studies have proven the need for therapeutic education among the patient, as the study of the Association of Life Supporting Organizations and Education for Diabetics (2010) (ASAVED) its sample consisted of 220 diabetic patients in two experimental and control groups. The therapeutic education program includes 7 sessions, which consisted of diet, definition of diabetes, physical activity, diabetic foot, and complications related to the disease, as

the program was presented during a period of 6 to 12 weeks at a rate of 2 hours in each session. The results were : A decrease in the rate of diabetic leukemia to the required extent, noticing changes in patients' behaviors to a positive trend towards their health, an increase in the level of good self-monitoring of diabetes compared to the control sample (Lekhal, 2010). Various attempts started in the therapeutic education among cancer patients as a result of the high incidence of cancer, Also changes in the relationship of caregivers to the patient have a role in the development of therapeutic education among cancer patients. As the study of J.Tavernier et al. (2012) entitled Therapeutic education and breast cancer : The effect of the pharmaceutical interview on the quality of life, this study indicated that breast cancer is one of the most common types of malignant cancers in the world, as it often decrease the quality of life among patients due to the steps of diagnosis and initial treatment, especially toxic chemotherapy. This study aimed to evaluate the effect of the pharmacist interview on improving the patients' quality of life, they compared quality of life in a retrospective study of two randomized groups among patients with breast cancer (regardless of stage of disease, treated with toxic chemotherapy), Quality of life was determined by a QLQ-C30 question related to EORTC, 20 patients received a pharmacist's interview during the first treatment, and 28 patients did not. The two groups shared ten items (age, new or adjuvant chemotherapy, TNM stage, SBR score, estrogen-progesterone receptor status, HER2 status, and chemotherapy cycle number). The results are that both

groups reported many side effects without differing in terms of occurrence frequency and type of side effects (nausea, constipation, diarrhea, inflammation of the oral mucosa). The group that received a pharmacist interview also showed a significant improvement in global quality of life compared to the group without an interview. Hence, this study shows the importance of the pharmacist participation or the pharmacist team in the development and improvement of therapeutic education programs in oncology (J.Tavernier & al , 2012). Experimental studies conducted mainly in the United States of America to evaluate the side effects of chemotherapy and pain management have proven that such educational programs can improve the quality of life for the patient and reduce the side effects of treatment. For example, the study by M.Sebyhan et al. (2021): Therapeutic Education Program at the forefront of cancer-related thrombosis care. Which aimed to treat anticoagulants associated with cancer (CAT), which requires specific approaches, although it is well regulated, as current national and international clinical practice guidelines (CPG) recommend the use of low molecular weight heparin over a period of 6 months as initial treatment. The program was applied to 182 cancer patients (average age 64.9 years), 53.3 percent of whom had metastatic (advanced) disease. The study also confirmed that anticoagulation should be maintained afterward because the cancer is active, a patient education program for CAT was established with the goal of improving the quality of care, a retrospective analysis was used for all patients who participated in the programme. The study concluded that

compliance improves when the patient understands his disease and his related to treatment.

Thus, It has been concluded that the success of such therapeutic education programs is linked to the individual (the patient), and this is due to its association with several criteria, including :

_ Taking in consideration the patient's opinion in the development and preparation of therapeutic education programmes.

_ Introducing teams specializing in therapeutic education and empowered with it, and having experience in various disciplines, to participate in educational work.

_ Using methodological tools to evaluate therapeutic education.

_ Attempting to carry out many and detailed studies on various parts of the country.

_ Providing therapeutic education for each type of cancer.

_ Improving, educating and explaining to doctors and members involved in therapeutic education the need for effective and positive communication with cancer patients to give a good result.

3. Conclusion :

Numerous studies have proven the role that therapeutic education plays in patients suffering from diabetes, asthma, blood pressure, cancer. It is considered the most dangerous disease, and its difficulty lies in not controlling its causes as well as the length of the period of health care needed

by patients, in addition to the changes it imposes on the lives of patients and their families in all respects. Which makes designing and building means and programs for therapeutic education and providing them with the first responsibilities of caregivers, especially those educational programs that aim to provide the necessary information about the disease and its treatments, and empowering them with the skills needed to improve their therapeutic adherence. As well as those aimed to improving communication between him and the various members of the caregivers team throughout the period of his disease, and perhaps even after it.

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